



Modelling Financial Markets

Chapter 1: Stylised Facts and Returns



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Acronyms used in this material

- ▣ ACF: AutoCorrelation Function
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARCH-LM: ARCH Lagrange Multiplier (test)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CLT: Central Limit Theorem
- ▣ CRIX: CRypto IndeX
- ▣ EGARCH: Exponential GARCH
- ▣ ETF: Exchange-Traded Fund
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GEO: Government Emergency Ordinance
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ JB: Jarque–Bera (test)
- ▣ LB: Ljung–Box (test)
- ▣ LM: Lagrange Multiplier
- ▣ MDD: Maximum DrawDown
- ▣ OHLC: Open, High, Low, Close
- ▣ SR: Sharpe Ratio
- ▣ US: United States

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What Is a Stylised Fact?

Definition

- ▣ A **stylised fact** is a statistical regularity observed across many assets, markets and periods
- ▣ It is qualitative, not quantitative: “returns have heavy tails”, not “kurtosis = 7.3”
- ▣ It describes **returns**, not price levels; the standard survey is [Cont \(2001\)](#)

Key message of this chapter

- ▣ The **direction** of returns is almost unpredictable, but their **size** (absolute value, a proxy for risk) is highly structured

Why Stylised Facts Matter

- They are the **benchmark** every model of returns must reproduce
- A model that violates them misprices risk: margins, capital and option prices are wrong
- The textbook model of normally distributed, i.i.d. returns fails almost all of them
 - ▶ i.i.d. = **independent and identically distributed**: every day is a new, independent draw from the same distribution
- Stylised facts tell us **which** features a better model needs
- Later chapters build on them: volatility (5, 6, 9), risk (7, 8), machine learning and bubbles (13, 17)

Data Used in This Chapter

Series	Market	Period
S&P 500 index	US equities (price index)	1990–2026
BET-TR	Romanian equities (total return)	2014–2026
Bitcoin	crypto-asset, traded 24/7	2014–2026
EUR/RON	exchange rate, managed float	Jul 2005–2026
Gold	spot price, USD per ounce	1990–2026

- Daily closing values up to 18 September 2026 (repository folder data/market)
- Unless stated otherwise, returns are daily log returns (defined in Section 2)

The Five Series

- ▣ **S&P 500**: Standard & Poor's index of 500 large US companies; a **price index**, dividends are not included
- ▣ **BVB**: the Romanian stock exchange
 - ▶ **BET**: the benchmark index of the BVB
 - ▶ **BET-TR**: its version with dividends re-invested
- ▣ **EUR/RON**: price of one euro in Romanian lei; a **managed float** (market rate with central-bank interventions)
 - ▶ Source: the daily reference rate of the **BNR**
- ▣ **Gold**: spot price in US dollars (USD) per troy ounce; weekday quotes only (weekend prints removed)
- ▣ **Bitcoin**: price in USD; trades 24 hours a day, 7 days a week (24/7)

Data Quality Notes

- EUR/RON: official BNR reference rate instead of exchange quotes
 - ▶ Exchange quote series showed lag-1 autocorrelation (correlation of consecutive daily changes) of -0.4 to -0.7 in several years
 - ▶ Such values come from quote errors, not from the market
- S&P 500 **OHLC** data: Open, High, Low and Close prices of each day
 - ▶ Open prices before 2007 equal the previous close: unusable for range-based estimators
 - ▶ Range estimators therefore start in 2008 (Section 6)
- Lesson: always inspect the data before computing statistics

Prices Before Computers: the Stock Ticker



Stock ticker designed by Thomas Edison
(Smithsonian collection)

Photo: Jaclyn Nash, Smithsonian NMAH; public
domain; Wikimedia Commons

- Before computers, trades reached brokers as prices printed on a paper **ticker tape**
- A daily close is one snapshot of this stream of prices; returns are built from such snapshots
- The choice of snapshot (close, fixing, 00:00 UTC) matters, as Chapter 0 showed
- Prices at the level of individual trades: market microstructure, Chapter 10

Why Returns, Not Prices?

Returns instead of prices

- ▣ **Scale-free:** a 1% move means the same for a \$10 and a \$1,000 stock
- ▣ Prices are **non-stationary**: they trend and wander without a fixed mean (a unit root)
- ▣ **Stationary:** the distribution does not change over time; returns are much closer to it
- ▣ Investors care about relative gains

Adjusted and total-return prices

- ▣ **Dividends:** the price drops on the ex-dividend date, wealth does not
- ▣ **Splits:** mechanical price jumps
- ▣ **Adjusted prices** correct past prices for both
- ▣ **Total return** series re-invest dividends (e.g. BET-TR); the gap to a price index grows each year by the dividend yield

Simple Returns

□ Notation: P_t price at the end of day t , D_t dividend paid in $(t-1, t]$

□ **One-period simple (net) return**

$$R_t = \frac{P_t + D_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1 \quad (\text{no dividends})$$

□ **Gross return:** $1 + R_t = P_t/P_{t-1}$, the factor by which wealth is multiplied

□ Simple returns are bounded below: $R_t \geq -100\%$ (limited liability)

□ Reference textbooks: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 1 and [Franke, Härdle & Hafner \(2019\)](#)

Multi-Period and Portfolio Simple Returns

- ▣ **Multi-period return** over k days: gross returns **multiply**

$$1 + R_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j})$$

- ▣ **Portfolio return**: simple returns aggregate **linearly** across assets

$$R_{p,t} = \sum_{i=1}^N w_i R_{i,t}, \quad \sum_i w_i = 1$$

► w_i : share of wealth invested in asset i at $t - 1$

- ▣ Advantage: exact portfolio accounting
- ▣ Disadvantage: products over time are awkward for statistics

Log Returns

- **Continuously compounded (log) return**

$$r_t = \ln(1 + R_t) = \ln P_t - \ln P_{t-1}$$

- Interpretation: the constant rate that, compounded continuously over one day, turns P_{t-1} into P_t
- Range: $r_t \in (-\infty, +\infty)$, convenient for Normal-distribution or Student- t models
- Back-transformation: $R_t = e^{r_t} - 1$

Properties of Log Returns

- **Time additivity:** multi-period log returns are sums

$$r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$$

- ▶ Sums of many terms allow CLT arguments
- ▶ CLT: a sum of many weakly dependent terms approximately follows the Normal distribution
- **Not additive across assets:** $r_{p,t} \neq \sum_i w_i r_{i,t}$; the approximation is good only for small returns
- Rule of thumb
 - ▶ Log returns for time-series modelling of one asset
 - ▶ Simple returns for portfolios and performance reporting

Worked Example: Small vs Large Moves

Simple return R	Log return $r = \ln(1 + R)$	Difference
+1%	+0.995%	0.005 pp
-1%	-1.005%	0.005 pp
+20%	+18.2%	1.8 pp
-20%	-22.3%	2.3 pp
+50%	+40.5%	9.5 pp
-50%	-69.3%	19.3 pp

- pp = percentage points: the difference between two percentages
- Daily moves are small, so $r \approx R$; for crashes and long horizons the two differ a lot
- For every $R \neq 0$: $\ln(1 + R) < R$ (equality only at $R = 0$): log returns are lower than simple returns

Worked Example: Adding Returns

Over time: +50% then -50%

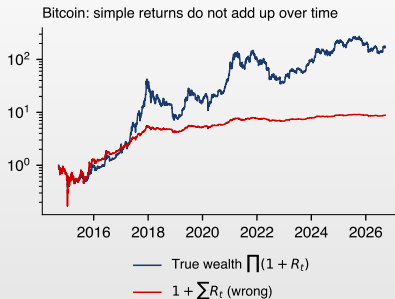
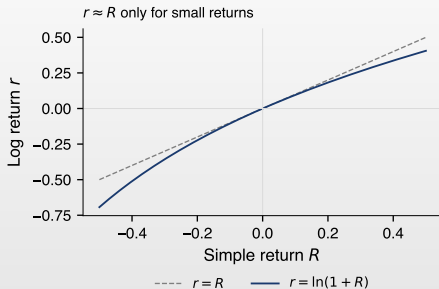
- ▣ Wealth: $1 \rightarrow 1.5 \rightarrow 0.75$, a loss of 25%
- ▣ Sum of simple returns: $0.5 - 0.5 = 0$ (misleading)
- ▣ Sum of log returns: $\ln 1.5 + \ln 0.5 = -0.288$
- ▣ $e^{-0.288} = 0.75$: exact

Across assets: 50/50 portfolio

- ▣ Asset A: $R = +10\%$; asset B: $R = -10\%$
- ▣ Portfolio simple return: $0.5(0.1) + 0.5(-0.1) = 0$: exact
- ▣ Weighted log returns:
 $0.5 \ln 1.1 + 0.5 \ln 0.9 = -0.005$
- ▣ True portfolio log return: $\ln 1 = 0$

- ▣ Log returns add over **time**; simple returns add over **assets**

Simple vs Log Returns in Practice



- Left: $r \approx R$ only for small moves; the gap grows with $|R|$
- Right: Bitcoin 2014–2026 multiplied wealth by 176.9, i.e. $\sum_t r_t = \ln 176.9 = 5.18$
- Adding daily simple returns gives $1 + \sum_t R_t = 8.83$: wrong by a factor of 20

Annualisation

- ▣ **Annualisation** converts daily statistics into yearly ones
- ▣ Let P be the number of periods per year
 - ▶ $P = 252$ trading days for stocks; $P = 365$ for crypto (traded every day); gold: its actual ≈ 260 weekday quotes a year
- ▣ If returns are i.i.d. with daily mean $\bar{r}$ and daily standard deviation $\hat{\sigma}$

$$\mu_{\text{ann}} = P \bar{r}, \quad \sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$$

- ▣ **Volatility** = standard deviation of returns, usually reported annualised
- ▣ The $\sqrt{P}$ rule fails under autocorrelation or volatility clustering (see Fact 6)

Worked Example: Annualising a Daily Volatility

S&P 500, 1990–2026

- Daily standard deviation $\hat{\sigma} = 1.133\%$
- $\sqrt{252} = 15.87$
- $\sigma_{\text{ann}} = 1.133\% \times 15.87 = 18.0\%$

Bitcoin, 2014–2026

- Daily standard deviation $\hat{\sigma} = 3.485\%$
- $\sqrt{365} = 19.10$
- $\sigma_{\text{ann}} = 3.485\% \times 19.10 = 66.6\%$

- Bitcoin is about 3.7 times as volatile as the S&P 500
- Using 252 instead of 365 for Bitcoin would understate its volatility by 17%

Arithmetic Mean, Geometric Mean and CAGR

- ▣ **Arithmetic mean:** the average one-period return, $\bar{R} = \frac{1}{T} \sum_t R_t$
- ▣ **Geometric mean:** the constant return that produces the same final wealth
- ▣ **CAGR** = Compound Annual Growth Rate: the geometric mean per year

$$\text{CAGR} = \left(\frac{P_T}{P_0} \right)^{1/Y} - 1, \quad Y = \text{number of years}$$

- ▣ Link between the two means

$$\mu_{\text{geo}} \approx \mu_{\text{arith}} - \frac{1}{2}\sigma^2$$

- ▣ The gap $\frac{1}{2}\sigma^2$ is the **volatility drag**: volatility lowers compound growth

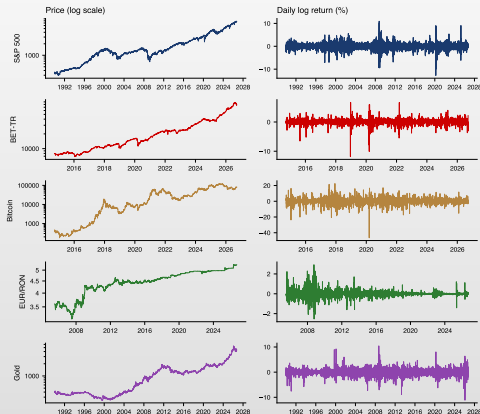
Worked Example: CAGR of the S&P 500

- Close on 2 January 1990: $P_0 = 359.69$; close on 18 September 2026: $P_T = 7,650.50$
- Wealth multiple: $P_T/P_0 = 21.27$
- Length: $Y = 36.71$ years
- $\text{CAGR} = 21.27^{1/36.71} - 1 = 8.69\%$ per year
- Check with log returns: $\ln 21.27/36.71 = 0.0833$, and $e^{0.0833} - 1 = 8.69\%$
- Price index only: with re-invested dividends the CAGR would be higher

Volatility Drag: Bitcoin

- Bitcoin 2014–2026, $P = 365$
 - ▶ Mean of daily simple returns $\times 365$: 65.2% per year (arithmetic)
 - ▶ Realised CAGR: 53.9% per year (geometric)
- The difference of 11 pp is the volatility drag of a very volatile asset
- Which one to report?
 - ▶ CAGR for realised, multi-year performance
 - ▶ Arithmetic mean for the expected return of the next period

Five Markets: Prices and Returns



- ▣ **Left:** prices on log scale, trending and non-stationary
- ▣ **Right:** daily log returns
 - ▶ Mean-reverting around zero
 - ▶ Calm and turbulent periods alternate
 - ▶ Extreme days cluster: 2008, March 2020
- ▣ EUR/RON moves more than 1% on only 1.4% of days
- ▣ Bitcoin, 12 March 2020: log return -46.5% , i.e. a price fall of $1 - e^{-0.465} = 37.2\%$

Recap: Prices and Returns

- Work with returns: scale-free and close to stationary
- Simple returns aggregate across assets; log returns aggregate over time
- Annualise with P and $\sqrt{P}$; use 365 for crypto
- CAGR (geometric) is below the arithmetic mean by about $\sigma^2/2$
- Use total-return or adjusted series when dividends matter

Mean, Variance and Volatility

- ▣ Sample of daily returns $r_1, \dots, r_T$
- ▣ **Sample mean**

$$\bar{r} = \frac{1}{T} \sum_{t=1}^T r_t$$

- ▣ **Sample variance and standard deviation**

$$\hat{\sigma}^2 = \frac{1}{T-1} \sum_{t=1}^T (r_t - \bar{r})^2, \quad \hat{\sigma} = \sqrt{\hat{\sigma}^2}$$

- ▣ Daily means are tiny (hundredths of a percent); daily standard deviations are 10–60 times larger in our data
- ▣ Consequence: the mean is estimated very imprecisely, the volatility much better

Skewness

- ▣ **Skewness** measures the asymmetry of a distribution

$$\hat{S} = \frac{1}{T} \sum_{t=1}^T \left(\frac{r_t - \bar{r}}{\hat{\sigma}} \right)^3$$

- ▣ Normal distribution: $S = 0$ (symmetric)
- ▣ $S < 0$ (negative skew): large losses are more frequent than equally large gains
- ▣ $S > 0$ (positive skew): large gains dominate, as for a lottery ticket
- ▣ Equity indices typically show negative skewness: crashes are sharper than rallies

Kurtosis and Excess Kurtosis

- ▣ **Kurtosis** measures the weight of the tails

$$\hat{K} = \frac{1}{T} \sum_{t=1}^T \left(\frac{r_t - \bar{r}}{\hat{\sigma}} \right)^4$$

- ▣ Normal distribution: $K = 3$; **excess kurtosis** = $K - 3$ (zero for the Normal distribution)
- ▣ $K > 3$ (**leptokurtosis**): more mass in the centre **and** in the tails than the Normal distribution
- ▣ Caution: $\hat{K}$ depends on the fourth power of returns
 - ▶ It is dominated by a few extreme days and converges very slowly
 - ▶ **Tail index** α : the exponent in $P(|X| > x) \approx c x^{-\alpha}$; if $\alpha < 4$ the population kurtosis does not exist

Autocorrelation and the ACF

- **Autocorrelation** at lag k : the correlation between r_t and r_{t-k}

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^T (r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^T (r_t - \bar{r})^2}$$

- **ACF** = AutoCorrelation Function: $\hat{\rho}_k$ plotted against $k = 1, 2, \dots$
- Under i.i.d. returns, $\hat{\rho}_k \approx N(0, 1/T)$
 - ▶ 95% band: $\pm 1.96/\sqrt{T}$; bars outside the band are significant
 - ▶ The band assumes i.i.d. returns: with heavy tails and volatility clustering it is too narrow (same issue as for Ljung-Box)
- We compute the ACF of r_t (linear predictability) and of $|r_t|$ or r_t^2 (predictability of volatility)

Hypothesis Tests: A Reminder

- H_0 (null hypothesis): the property we test, e.g. “returns follow the Normal distribution”
- **Test statistic**: a number computed from the data, large when H_0 looks false
- **Critical value**: the threshold that the statistic exceeds with probability 5% if H_0 is true
- **p-value**: the probability, under H_0 , of a statistic at least as large as the observed one
 - ▶ Reject H_0 at the 5% level if p-value < 0.05
- χ_m^2 (chi-square with m degrees of freedom): the reference distribution of the three tests below
 - ▶ 5% critical values: $\chi_2^2 = 5.99$, $\chi_5^2 = 11.1$, $\chi_{10}^2 = 18.3$

The Jarque–Bera Test

- ▣ **JB** = Jarque–Bera test of normality, [Jarque & Bera \(1980\)](#)
- ▣ H_0 : returns follow the Normal distribution, i.e. skewness 0 and excess kurtosis 0

$$JB = \frac{T}{6} \left(\hat{S}^2 + \frac{(\hat{K} - 3)^2}{4} \right) \stackrel{H_0}{\sim} \chi^2_2$$

- ▣ Large skewness or excess kurtosis give a large JB
- ▣ With thousands of observations, even small deviations from normality are detected

Worked Example: JB for the S&P 500

- ▣ S&P 500, 1990–2026: $T = 9,245$, $\hat{S} = -0.364$, $\hat{K} - 3 = 10.88$
- ▣ Skewness term: $\hat{S}^2 = 0.132$
- ▣ Kurtosis term: $10.88^2/4 = 29.6$
- ▣ $JB = \frac{9245}{6} (0.132 + 29.6) = 1540.8 \times 29.73 \approx 45,800$
- ▣ Critical value 5.99: normality is rejected by a factor of about 7,600
- ▣ Almost all of the statistic comes from the kurtosis term: the problem is the tails, not the asymmetry

The Ljung–Box Test

- ▣ **Ljung–Box (LB)** test of no autocorrelation up to lag m , [Ljung & Box \(1978\)](#)
- ▣ $H_0: \rho_1 = \dots = \rho_m = 0$

$$Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k} \stackrel{H_0}{\sim} \chi_m^2$$

- ▣ Applied to r_t : linear predictability of returns
- ▣ Applied to r_t^2 : predictability of volatility
- ▣ We use $m = 10$ lags (critical value 18.3)

The ARCH-LM Test

- ▣ **ARCH** = AutoRegressive Conditional Heteroskedasticity, Engle (1982)
 - ▶ Conditional heteroskedasticity: the variance of today's return depends on past returns
- ▣ **LM** = Lagrange Multiplier: the testing principle used
- ▣ Procedure with q lags (we use $q = 5$)
 - ▶ Regress $\hat{\varepsilon}_t^2$ on $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-q}^2$, where $\hat{\varepsilon}_t = r_t - \bar{r}$
 - ▶ Statistic: $TR^2 \sim \chi_q^2$ under H_0 of no ARCH effects
- ▣ A large statistic means that volatility is predictable from past squared returns

Summary Statistics: Five Markets

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Observations	9245	2992	4384	5352	9559
Mean (% per day)	0.033	0.077	0.118	0.007	0.025
Std. dev. (% per day)	1.13	0.98	3.49	0.28	1.01
Annualised volatility (%)	18.0	15.5	66.6	4.4	16.3
Skewness	-0.36	-1.40	-0.70	0.73	-0.23
Excess kurtosis	10.9	17.8	12.0	15.1	9.3
Min / max log return (%)	-12.8 / 11.0	-11.9 / 6.7	-46.5 / 22.5	-2.5 / 2.9	-11.1 / 10.4
Jarque-Bera	45805	40433	26482	51575	34732
Ljung-Box $Q(10), r_t$	91.6	70.0	20.2	221.7	32.1
Ljung-Box $Q(10), r_t^2$	8018	714	213	2420	1049
ARCH-LM (5 lags)	2229	371	115	871	483

□ Min and max are daily **log** returns; EUR/RON from 1 July 2005 (new leu, RON)

□ 5% critical values: $\chi_2^2 = 5.99$, $\chi_{10}^2 = 18.3$, $\chi_5^2 = 11.1$; all p-values < 0.03

Reading the Table

- ▣ **Normality is rejected everywhere:** JB statistics in the tens of thousands versus a critical value of 6
- ▣ **Heavy tails:** excess kurtosis between 9.3 (gold) and 17.8 (BET-TR)
- ▣ **Negative skewness** for equity, crypto and gold; strongest for BET-TR (-1.40)
- ▣ **Volatility is predictable:** $Q(10)$ on r_t^2 is 10–90 times larger than on r_t ; ARCH-LM rejects everywhere
- ▣ **Linear autocorrelation** of r_t : statistically significant but economically small
 - ▶ Large samples make tiny correlations significant; heavy tails and clustering distort the standard test
- ▣ EUR/RON is special: tiny volatility, high kurtosis, positive skewness, strong autocorrelation (Section 5)

Recap: Descriptive Statistics and Tests

- ▣ Skewness and excess kurtosis summarise asymmetry and tail weight
- ▣ JB tests normality; Ljung–Box tests autocorrelation; ARCH-LM tests predictable volatility
- ▣ All five markets reject normality and constant variance
- ▣ Statistical significance is not economic significance in samples of thousands of days

The People Behind the Stylised Facts (1/2)



Carl Friedrich Gauss (1777–1855)

Painting: C.A. Jensen (1840); public domain;
Wikimedia Commons

- The Normal distribution: the benchmark that returns violate



Louis Bachelier (1870–1946)

Photo: unknown author; public domain;
Wikimedia Commons

- Prices as a random walk, in his thesis Bachelier (1900)

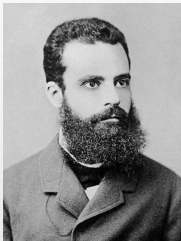


William S. Gosset, “Student” (1876–1937)

Photo: unknown author; public domain;
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- The Student- t distribution Student (1908), today a model for heavy tails

The People Behind the Stylised Facts (2/2)



Vilfredo Pareto (1848–1923)

Photo: unknown author (1870s); public domain; [Wikimedia Commons](#)



Benoît Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 FR; [Wikimedia Commons](#)

- Pareto: **power laws** for large incomes, the ancestor of heavy-tail models
- Mandelbrot: cotton prices have heavy tails and the Normal-distribution model fails [Mandelbrot \(1963\)](#)
- Their tail index α is what the Hill estimator in this chapter measures

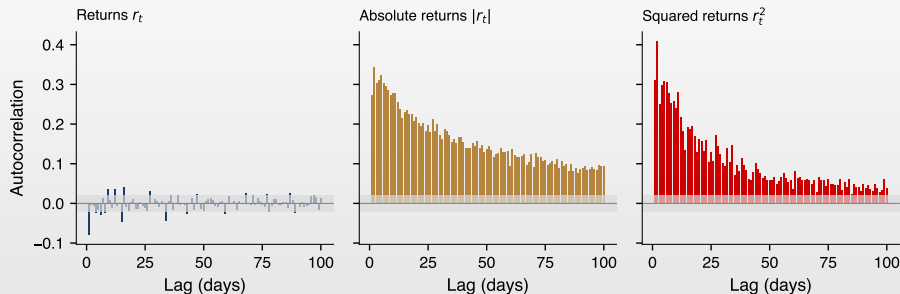
The Stylised Facts of Cont (2001)

1. Absence of linear autocorrelation
 2. Heavy tails
 3. Gain/loss asymmetry
 4. Aggregational Gaussianity
 5. Intermittency
 6. Volatility clustering
 7. Conditional heavy tails
 8. Slow decay of autocorrelation in absolute returns
 9. Leverage effect
 10. Volume/volatility correlation
 11. Asymmetry in time scales
- We verify most of them on the five markets, with data up to September 2026
 - Early evidence: Mandelbrot (1963), Fama (1965); textbook treatment: Taylor (2008), Franke, Härdle & Hafner (2019)

Facts 1 and 6: Uncorrelated but Not Independent

- **Fact 1 — absence of autocorrelation:** $\text{corr}(r_t, r_{t-k}) \approx 0$ for $k \geq 1$
 - ▶ Past returns do not predict the direction of future returns linearly
- **Fact 6 — volatility clustering:** large moves tend to be followed by large moves, of either sign
 - ▶ $\text{corr}(|r_t|, |r_{t-k}|) > 0$ for many lags
- Together: returns are **uncorrelated but not independent**
- Independence would require zero correlation of **every** function of returns, including $|r_t|$

Facts 1 and 6 on the S&P 500



- ▣ S&P 500 1990–2026, lags 1–100; grey band: $\pm 1.96/\sqrt{T}$
- ▣ Returns: almost all autocorrelations inside the band (lag 1: -0.08)
- ▣ Absolute and squared returns: large, positive and persistent (lag 1: 0.27 for $|r_t|$)

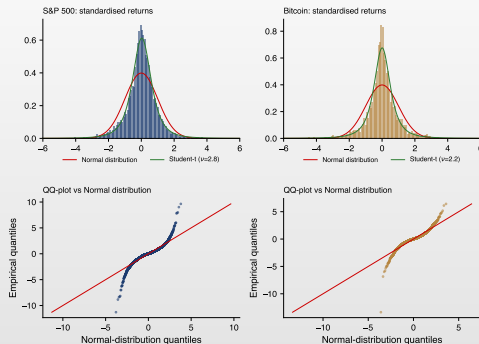
Why Facts 1 and 6 Matter for Modelling

- Fact 1: simple linear forecasts of returns (e.g. AR models) have little value
 - ▶ AR = autoregressive model: r_t regressed on its own past values
- Fact 6: the **variance** must be modelled as time-varying
 - ▶ $r_t = \sigma_t \varepsilon_t$ with σ_t depending on past returns
 - ▶ This is the idea behind GARCH models (Chapter 5)
- Risk forecasts (margins, capital requirements) must be updated as volatility changes

Fact 2: Heavy Tails

- ▣ **Heavy (fat) tails:** extreme returns occur far more often than under a Normal distribution with the same variance
- ▣ Diagnostic tools
 - ▶ Excess kurtosis above zero
 - ▶ Histogram compared with the density of the Normal distribution
 - ▶ **QQ-plot** (quantile–quantile plot): empirical quantiles against quantiles of the Normal distribution
- ▣ Reading a QQ-plot: points on the diagonal = Normal distribution; points bending away in the ends = heavy tails
- ▣ **Quantile** q_α : the value below which a fraction α of the data falls

Heavy Tails: Histograms and QQ-Plots



- Standardised returns $(r_t - \bar{r})/\hat{\sigma}$: higher peak and fatter tails than the Normal distribution
- A Student- t fits much better (next slides); QQ-plots bend away from the line in both tails

How Heavy Are the Tails?

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Days beyond $3\hat{\sigma}$ (%)	1.55	1.44	1.82	2.06	1.42
Days beyond $5\hat{\sigma}$ (count)	30	12	10	29	26
Expected under the Normal distribution, $> 5\hat{\sigma}$	0.005	0.002	0.003	0.003	0.005
Student- t degrees of freedom $\hat{\nu}$ (MLE fit)	2.8	3.1	2.2	1.2	2.9

- Normal distribution: 0.27% of days beyond 3σ ; observed 5–8 times more
- Normal distribution: one 5σ day every 6,900 years of trading; observed 10–30 in 12–36 years
- $\hat{\nu}$: Student- t degrees of freedom (next slide), estimated by **maximum likelihood**: the value that makes the observed returns most probable

The Student- t Distribution

- Density with ν degrees of freedom, location μ and scale s

$$f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\nu\pi} s} \left(1 + \frac{(x - \mu)^2}{\nu s^2}\right)^{-\frac{\nu+1}{2}}$$

- ▶ $\Gamma(\cdot)$: the gamma function, a continuous extension of the factorial
- ν controls the tails: small ν = heavy tails
- $\nu \rightarrow \infty$ gives the Normal distribution; $\nu = 1$ gives the Cauchy distribution
- Student- t as a model of returns: [Praetz \(1972\)](#)

Student- t : Tails, Moments and Our Estimates

- ▣ Tails decay as a **power law**: $P(X > x) \propto x^{-\nu}$ for large x
 - ▶ Much slower than the tail of the Normal distribution, which decays like $e^{-x^2/2}$
- ▣ Moments exist only up to order $< \nu$
 - ▶ Variance $s^2\nu/(\nu - 2)$ for $\nu > 2$; excess kurtosis $6/(\nu - 4)$ for $\nu > 4$
- ▣ Our fits give $\hat{\nu} \approx 1.2$ – 3.1 , but $\hat{\nu}$ is **not** a tail-index estimator
 - ▶ The likelihood is dominated by the many small returns in the centre, not by the tail
 - ▶ EUR/RON $\hat{\nu} = 1.2$ is an artefact of the fixing: 57% of daily changes are below 0.1%, plus rare jumps
- ▣ The tail itself is estimated with the **Hill estimator** (next slides)
- ▣ Power-law tails were first argued by [Mandelbrot \(1963\)](#) and [Fama \(1965\)](#)

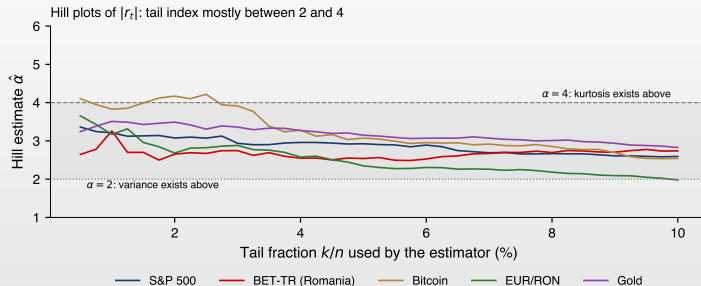
The Hill Estimator of the Tail Index

- ▣ **Tail index** α : $P(|X| > x) \approx c x^{-\alpha}$ for large x ; moments of order $< \alpha$ exist
 - ▶ $\alpha > 2$: finite variance; $\alpha > 4$: finite kurtosis
- ▣ **Hill estimator**, [Hill \(1975\)](#): order the absolute returns $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)}$ and use the k largest

$$\hat{\alpha}_k = \left[\frac{1}{k} \sum_{i=1}^k \ln X_{(i)} - \ln X_{(k+1)} \right]^{-1}$$

- ▣ **Choice of k** : a bias–variance trade-off
 - ▶ Small k : few observations, noisy; large k : the centre of the distribution enters, biased
 - ▶ **Hill plot**: $\hat{\alpha}_k$ against k ; read the level where the curve is flat (here $k = 1\text{--}5\%$ of n)

Hill Plots for Five Markets



- $\hat{\alpha}$ for $|r_t|$ at $k = 2.5\%$ of n : S&P 500 3.1, BET-TR 2.7, Bitcoin 4.2, EUR/RON 2.8, gold 3.3
- For $k = 1\text{--}5\%$: $\hat{\alpha}$ between 2.3 and 3.5 except Bitcoin (3.1–4.2); loss tails heavier for BET-TR (2.2 vs 3.1) and Bitcoin (2.7 vs 3.4)
- Variance exists, but the **fourth moment very likely does not**: sample kurtosis never settles; tails are modelled with extreme value theory (Chapter 7)

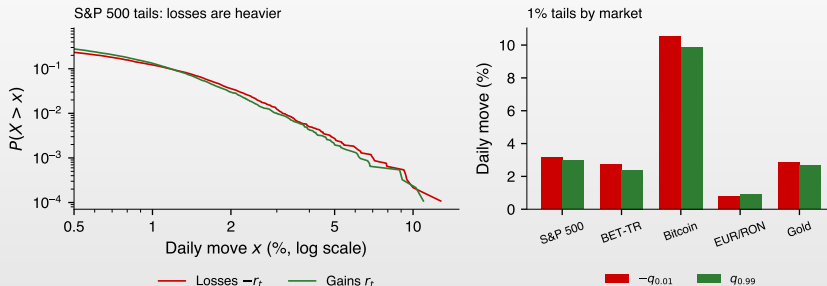
Why Heavy Tails Matter for Modelling

- ▣ **VaR** = Value-at-Risk: the loss exceeded with a small probability (e.g. 1%) over a horizon
 - ▶ It is a quantile of the loss distribution (Chapter 7)
- ▣ A Normal-distribution model with the right variance **underestimates** extreme quantiles
- ▣ Stress scenarios based on “ 3σ events” are far too optimistic
- ▣ Estimators based on high moments (kurtosis) are unstable: prefer quantiles and tail counts
- ▣ Remedies: Student- t or skewed distributions, extreme value theory (Chapter 7)

Fact 3: Gain/Loss Asymmetry

- **Statement:** large losses are more frequent or larger than gains of the same probability
- **Tools**
 - ▶ **Survival function:** $P(X > x)$, the probability of a move larger than x
 - ▶ Compare it for losses ($-r_t$) and gains (r_t) on a log-log scale
 - ▶ Compare the 1% quantile ($q_{0.01}$, losses) with the 99% quantile ($q_{0.99}$, gains)
- Holds for equity indices; not for all asset classes

Gain/Loss Asymmetry in the Data

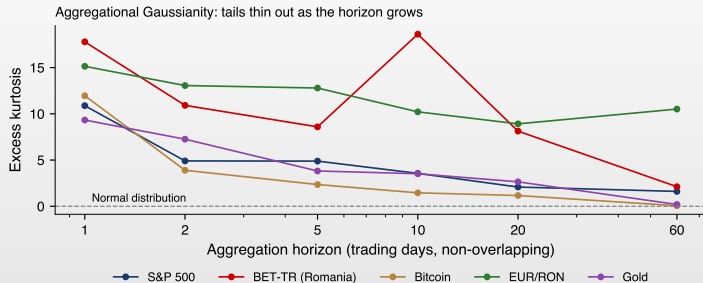


- The 1% loss quantile exceeds the 99% gain quantile for 4 of 5 markets
 - S&P 500: 3.15% vs 2.97%; BET-TR: 2.73% vs 2.40%; Bitcoin: 10.5% vs 9.9%
- EUR/RON is the exception: leu depreciations (up-moves) are larger (0.94% vs 0.76%)

Fact 4: Aggregational Gaussianity

- **Statement:** as the horizon grows, returns get closer and closer to the Normal distribution
- h -day log return: $r_t(h) = r_t + r_{t-1} + \cdots + r_{t-h+1}$
 - ▶ We use non-overlapping blocks, so that the h -day returns do not share days
- Mechanism: the CLT applies to sums of many daily returns
- Convergence is slow because daily returns are dependent (volatility clustering)

Aggregational Gaussianity in the Data



- Excess kurtosis of h -day returns, $h = 1, \dots, 60$
- Tails thin out: S&P 500 from 10.9 (daily) to 1.6 ($h = 60$), Bitcoin from 12.0 to 0.1
- $h = 60$ is a quarter for the S&P 500 (252 days a year) but two months for Bitcoin (365 days a year)
- At $h = 60$ only 154 (S&P 500) and 73 (Bitcoin) non-overlapping returns remain: approximate standard error $\sqrt{24/n} = 0.39$ and 0.57 under normality, so these values are very imprecise

Why Heavy Tails? Mixtures of Normals

- ▣ Suppose $r_t = \sigma_t \varepsilon_t$ with $\varepsilon_t \sim N(0, 1)$ independent of a random volatility σ_t

$$K = \frac{\mathbb{E}[r_t^4]}{\mathbb{E}[r_t^2]^2} = 3 \frac{\mathbb{E}[\sigma_t^4]}{\mathbb{E}[\sigma_t^2]^2} \geq 3$$

- ▣ Jensen's inequality ($\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$) with $X = \sigma^2$: any variation in volatility creates excess kurtosis
- ▣ Example: $\sigma = 1\%$ on 90% of days and 3% on 10% of days

$$K = 3 \cdot \frac{0.9 \cdot 1 + 0.1 \cdot 81}{(0.9 + 0.1 \cdot 9)^2} = 3 \cdot \frac{9.0}{3.24} = 8.33$$

- ▣ This links heavy tails (Fact 2) to volatility clustering (Fact 6)

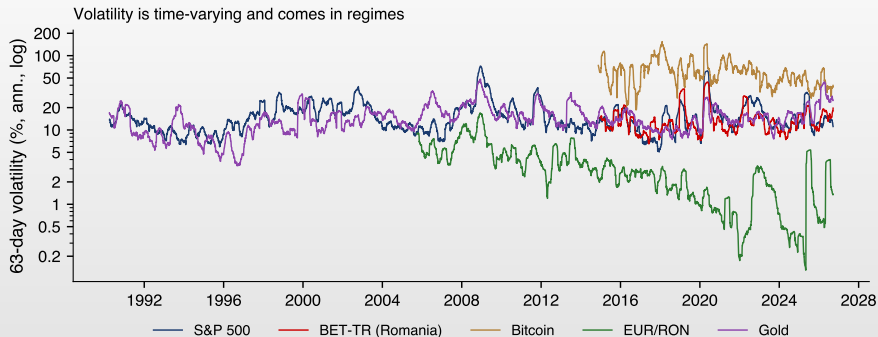
Fact 5: Intermittency

- ▣ **Statement:** returns show bursts of high volatility separated by calm periods (regimes)
- ▣ **Tool:** **rolling volatility** over a window of n days

$$\hat{\sigma}_t^{(n)} = \sqrt{P} \sqrt{\frac{1}{n-1} \sum_{j=0}^{n-1} \left(r_{t-j} - \bar{r}_t^{(n)} \right)^2}$$

- ▶ $\bar{r}_t^{(n)}$: mean over the same window; we use $n = 63$ days (one quarter)
- ▣ Constant-volatility models cannot produce such bursts

Intermittency in the Data

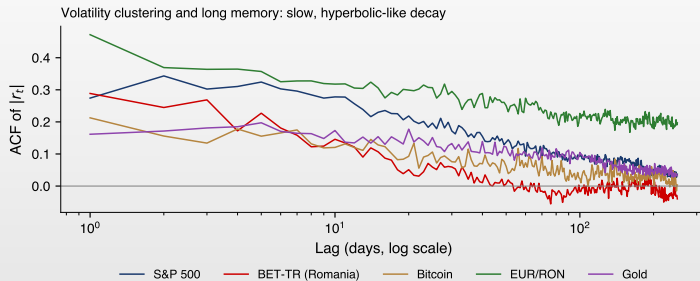


- 63-day rolling volatility, annualised, log scale
- Common spikes across markets: 2008–2009, March 2020
- EUR/RON volatility collapsed after 2012: an exchange-rate regime managed by the central bank

Fact 8: Slow Decay — Long Memory in Volatility

- ▣ **Statement:** the ACF of $|r_t|$ stays positive for months
- ▣ **Short memory:** autocorrelations decay **exponentially**, $\rho_k \approx c \phi^k$ with $|\phi| < 1$
 - ▶ Example: GARCH(1,1) volatility (Chapter 5)
- ▣ **Long memory:** autocorrelations decay **hyperbolically**, $\rho_k \approx c k^{-\beta}$ with $0 < \beta < 1$
 - ▶ Shocks to volatility die out very slowly
- ▣ Evidence: [Ding, Granger & Engle \(1993\)](#)

Long Memory in the Data



- ACF of $|r_t|$ up to one year (250 lags), log scale on the lag axis
- S&P 500: 0.27 at lag 1, 0.22 at lag 20, still 0.09 at lag 100
- BET-TR decays fastest (shorter sample); decay is slower than the exponential decay of GARCH(1,1)

Infinite Fourth Moments and Statistics on r_t^2

- ▣ The ACF of r_t^2 is a ratio of fourth-order moments: it needs $E[r_t^4] < \infty$, i.e. tail index $\alpha > 4$
- ▣ Our Hill estimates are about 2.5–3.5: the fourth moment is very likely infinite
 - ▶ The sample ACF of r_t^2 then converges slowly, to a limit that is not the Normal distribution; the $\pm 1.96/\sqrt{T}$ band is not valid, [Davis & Mikosch \(1998\)](#)
 - ▶ Ljung–Box on r_t^2 and ARCH-LM inherit the problem: the statistics are huge, but their p-values are not reliable
- ▣ The conclusion “volatility clusters” is robust; the exact magnitudes are not
- ▣ Remedy: use $|r_t|$, whose ACF needs only $\alpha > 2$ — this is one reason behind the Taylor effect (next slide)

The Taylor Effect

- **Definition:** compute the autocorrelation of $|r_t|^\delta$ for different powers $\delta > 0$
 - ▶ $\delta = 1$: absolute returns; $\delta = 2$: squared returns
- **Taylor effect:** the autocorrelation is highest for δ around 1, not for $\delta = 2$
- Implication: absolute returns are a more persistent, more robust measure of volatility than squared returns
- Documented by Taylor (1986); the name “Taylor effect” was given by Granger & Ding (1995); see also Ding, Granger & Engle (1993)

The Taylor Effect in the Data



- S&P 500: autocorrelation of $|r_t|^\delta$ as a function of δ
- Maximum at $\delta = 1.75$ (lag 1), 1.5 (lag 5) and 1.25 (lag 20): the optimum moves towards 1 at longer lags
- Motivates power-GARCH models and absolute-value volatility measures

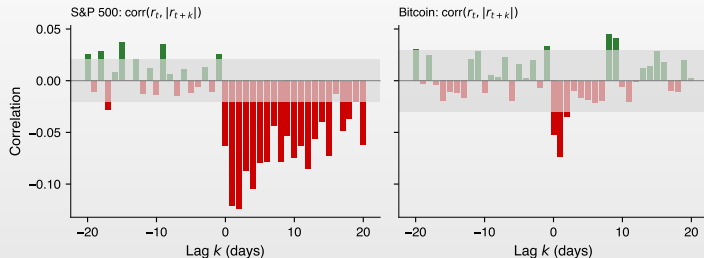
Fact 9: The Leverage Effect

- ▣ **Statement:** falling prices raise **future** volatility more than rising prices
- ▣ **Leverage correlation** at lag k

$$L(k) = \text{corr}(r_t, |r_{t+k}|)$$

- ▶ $L(k) < 0$ for $k > 0$: a negative return today is followed by larger moves
- ▶ $L(k) \approx 0$ for $k < 0$: past volatility does not predict today's sign
- ▣ Explanations: financial leverage of firms, [Christie \(1982\)](#); the retarded volatility model, [Bouchaud, Matacz & Potters \(2001\)](#)

The Leverage Effect in the Data

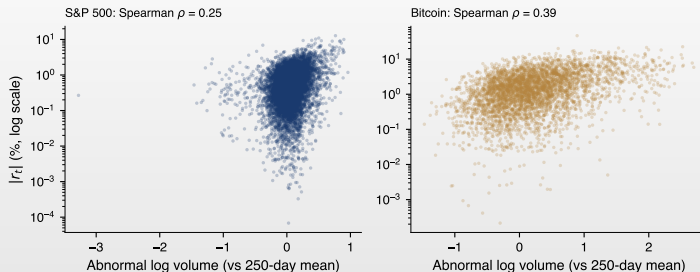


- S&P 500: $L(1) = -0.12$, still -0.07 at $k = 10$; no effect for $k < 0$
- BET-TR -0.12 ; weak for Bitcoin (-0.07), negligible for gold (-0.02), reversed for EUR/RON ($+0.10$)
- Modelling: asymmetric volatility models, GJR-GARCH (Glosten–Jagannathan–Runkle) and EGARCH (Exponential GARCH), Chapter 5

Fact 10: Volume and Volatility Move Together

- ▣ **Statement:** days with large price moves are days with high trading volume
- ▣ **Abnormal log volume:** $\ln V_t$ minus its 250-day moving average
- ▣ **Spearman rank correlation:** the correlation between the ranks of two variables
 - ▶ Robust to outliers and to non-linear but monotone relations
- ▣ Mixture-of-distributions hypothesis, [Clark \(1973\)](#): prices move in a random “business time” driven by information flow, which also drives volume
- ▣ Survey of the price–volume relation: [Karpoff \(1987\)](#)

Volume and Volatility in the Data



- Abnormal log volume against $|r_t|$
- Spearman rank correlation: 0.25 for the S&P 500 and 0.39 for Bitcoin
- Volume is a useful additional input for volatility forecasts
- Bitcoin caveat: volume is exchange-reported and aggregated over many venues; part of it is **wash trading** (an exchange trading with itself to inflate volume), [Cong, Li, Tang & Yang \(2023\)](#)

Facts 7 and 11: Conditional Tails and Time Scales

Fact 7: conditional heavy tails

- After removing volatility clustering, standardised returns are still heavy-tailed
- Fit a volatility model, e.g. GARCH (Chapter 5), get $\hat{\sigma}_t$, compute $\hat{z}_t = r_t / \hat{\sigma}_t$
- Kurtosis of $\hat{z}_t$: lower than for r_t , still above 3
- Models need Student- t or skewed innovations

- Both facts are documented in [Cont \(2001\)](#)

Fact 11: asymmetry in time scales

- Weekly volatility predicts daily volatility better than the reverse
- Traders with different horizons react at different frequencies
- **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility (Chapter 9)

Recap: The Stylised Facts

- ▣ Uncorrelated returns, strongly dependent volatility (Facts 1, 6, 8)
- ▣ Heavy tails, also conditionally, that thin out slowly with aggregation (Facts 2, 4, 7)
- ▣ Crashes sharper than rallies in equity markets (Facts 3, 9)
- ▣ Volatility in regimes, linked to volume and information flow (Facts 5, 10)
- ▣ Different horizons interact asymmetrically (Fact 11)

Crypto-Assets: The Same Facts, Amplified

Bitcoin vs S&P 500

- ▣ Volatility 66.6% vs 18.0% per year
- ▣ Excess kurtosis 12.0 vs 10.9
- ▣ Skewness -0.70 vs -0.36
- ▣ Worst day, log return: -46.5% (12 March 2020) vs -12.8% (16 March 2020)
- ▣ As price changes: -37.2% vs -12.0%

But some facts are weaker

- ▣ Leverage effect: weak, short-lived
- ▣ Volatility memory: ACF of $|r_t|$ decays faster (lag 100: 0.02 vs 0.09)
- ▣ Trading 24/7: 365 periods per year

Crypto-Assets: An Asset Class of Their Own

- Crypto as an asset class with its own statistical profile: [Härdle, Harvey & Reule \(2020\)](#)
- Tail, memory and moment factors classify cryptos as a separate asset class, mainly through the tail factor: [Pele et al. \(2023\)](#)
- Crypto modelling and the CRIX index in quantitative finance: [Härdle, Chen & Overbeck \(2017\)](#)
 - ▶ CRIX = CRypto IndeX, a benchmark index of the crypto market
- Deeper treatment in Chapter 16 (digital assets)

EUR/RON: A Managed Exchange Rate

- ▣ **Managed float:** the exchange rate is market-determined, but the central bank intervenes to limit fluctuations
- ▣ BNR reference rate, 1 July 2005–2026: from the new leu (RON, 1 RON = 10,000 old lei), as in Chapter 0
 - ▶ **Low volatility** (4.4% per year) but **high kurtosis** (15.1)
 - ▶ Long calm periods interrupted by sudden adjustments
 - ▶ Largest daily moves: October 2008 (+2.9% on 3 October, −2.5% on 9 October)
- ▣ **Positive** lag-1 autocorrelation (0.17), unlike equities
 - ▶ The central bank smooths adjustments over several days; a daily fixing, not a traded price

EUR/RON: Lessons

- Stylised facts are **market-specific**: always check the institutional setting
- Low volatility does not mean low tail risk: kurtosis is the second highest of the five series
- Data source matters
 - ▶ Exchange quote series showed spurious negative autocorrelation (quote errors)
 - ▶ Official reference rates avoid these errors

Romania: The BET Total Return Index

- ▣ BET-TR, 2014–2026, 2,992 days
- ▣ Facts confirmed: excess kurtosis 17.8, clustering ($Q(10)$ on r_t^2 : 714), 1% loss quantile 2.7%
- ▣ Strongest negative skewness of the five series (-1.40)
 - ▶ Worst day (log return) -11.9% on 19 December 2018: reaction to the bank-asset tax announced by the government on 18 December
 - ▶ The tax was adopted on 28 December 2018 as Government Emergency Ordinance (GEO) 114/2018
 - ▶ Policy shocks produce single-day crashes without comparable rallies

Romania: Autocorrelation in a Smaller Market

- ▣ Positive lag-1 autocorrelation (0.06), in contrast with the S&P 500 (-0.08)
- ▣ **Non-synchronous trading:** index constituents do not all trade at every moment
 - ▶ The index reflects news with a delay, creating positive autocorrelation, [Campbell, Lo & MacKinlay \(1997\)](#), ch. 3
- ▣ Frontier and emerging markets (below developed status in index-provider classifications)
 - ▶ Lower liquidity (it is costlier to trade quickly)
 - ▶ Stronger autocorrelation and policy-driven jumps

Why Use Open, High, Low and Close?

- **Close-to-close** estimator: the standard deviation of daily close-to-close log returns
 - ▶ Uses one number per day and ignores intraday information
- **Range-based** estimators use the OHLC prices
 - ▶ Notation: o_t, h_t, l_t, c_t are the log open, high, low and close of day t
 - ▶ The range $h_t - l_t$ is large on volatile days even if the close equals the open
- **Efficiency**: the ratio of variances of two estimators; higher efficiency = less noise for the same sample

Parkinson and Garman–Klass Estimators

- ▣ **Parkinson**, [Parkinson \(1980\)](#): uses only the daily range

$$\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \frac{1}{n} \sum_{t=1}^n (h_t - l_t)^2$$

- ▣ **Garman–Klass**, [Garman & Klass \(1980\)](#): combines the range with the open-to-close move

$$\hat{\sigma}_{GK}^2 = \frac{1}{n} \sum_{t=1}^n \left[\frac{1}{2} (h_t - l_t)^2 - (2 \ln 2 - 1) (c_t - o_t)^2 \right]$$

- ▣ Derived for a Brownian motion without drift and without overnight jumps
 - ▶ Brownian motion: the continuous-time random walk (Chapter 11)
 - ▶ Efficiency relative to close-to-close: about 5 (Parkinson) and 7–8 (Garman–Klass)

Rogers–Satchell and Yang–Zhang Estimators

- **Rogers–Satchell**, Rogers & Satchell (1991): unbiased under a non-zero **drift** (expected trend of the log price)

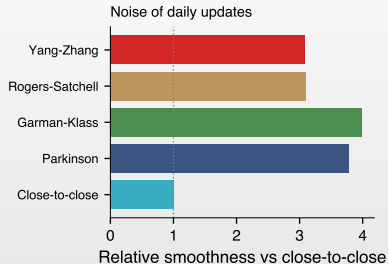
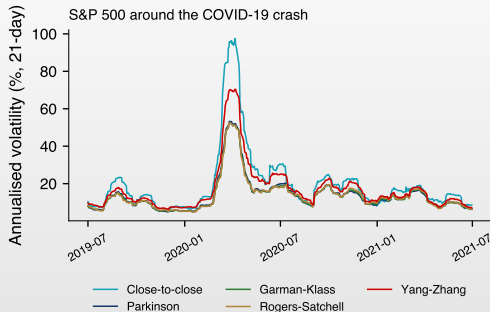
$$\hat{\sigma}_{RS}^2 = \frac{1}{n} \sum_{t=1}^n [(h_t - o_t)(h_t - c_t) + (l_t - o_t)(l_t - c_t)]$$

- **Yang–Zhang**, Yang & Zhang (2000): adds the overnight (close-to-open) jump

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_{\text{overnight}}^2 + k \hat{\sigma}_{\text{open-close}}^2 + (1 - k) \hat{\sigma}_{RS}^2, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

- ▶ $\hat{\sigma}_{\text{overnight}}^2$: variance of $o_t - c_{t-1}$; $\hat{\sigma}_{\text{open-close}}^2$: variance of $c_t - o_t$
 - ▶ Minimum-variance combination among drift-independent estimators robust to opening jumps
- Range estimators are the bridge to realised volatility from intraday data (Chapter 9)

Estimators on the S&P 500



- 21-day windows, 2008–2026; averages: close-to-close 16.4%, Yang–Zhang 13.9%, Parkinson 13.0%, Garman–Klass 12.4%, Rogers–Satchell 12.3%
- Pure range estimators are lower: they miss overnight gaps
- Yang–Zhang includes the overnight gap but stays 2.5 pp below close-to-close: the index open is partly stale (next slide)
- Range-based estimates are 3–4 times smoother (less noise in daily updates)

Data Quality: Stale Opens of the S&P 500 Index

- ▣ The index **open** is computed from the constituents' first prices; at 9:30 many stocks have not traded yet, so their previous close enters the open
- ▣ Open = previous close: about 96% of days until 2005, 3.5% over 2008–2026 (up to 26% in 2012–2013); on other days part of the open is still stale
- ▣ Comparison 2008–2026 with the SPY ETF, whose open is a traded price (annualised, %)

	Close-to-close	Yang–Zhang	Overnight	Open-to-close
S&P 500 index	16.4	13.9	6.7	17.3
SPY ETF	16.2	16.8	12.0	15.2

- ▣ A stale open moves part of the overnight move into the day: the index overnight volatility is understated and Yang–Zhang falls below close-to-close
- ▣ Lesson: for OHLC estimators use a traded instrument (ETF, futures), not the index

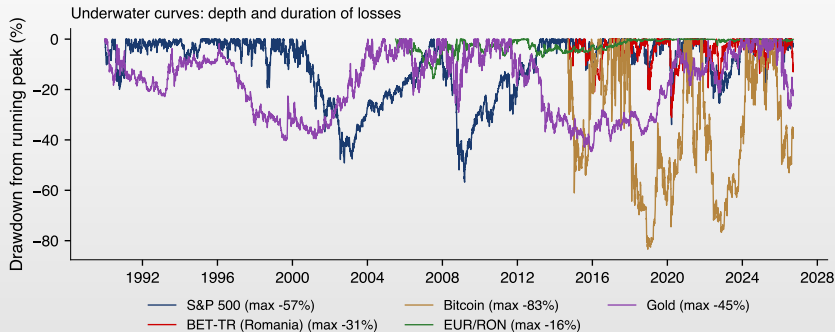
Drawdowns: Reminder and Focus

- Drawdown DD_t and maximum drawdown MDD were defined in Chapter 0 (section Key Measures)
- Here we add two things
 - ▶ A comparison of depth across the five markets
 - ▶ **Recovery duration**: the time from the peak until the price regains it
- Investors feel drawdowns more than volatility: a -50% loss needs $+100\%$ to recover

Worked Example: The S&P 500 in 2007–2009

- ▣ Peak: 9 October 2007, close 1,565.15
- ▣ Trough: 9 March 2009, close 676.53
- ▣ Maximum drawdown: $676.53/1565.15 - 1 = -56.8\%$
- ▣ Gain needed to recover: $1565.15/676.53 - 1 = +131\%$
- ▣ The price index regained the 2007 peak only on 28 March 2013, five and a half years after the peak

Drawdowns Across Markets



- Maximum drawdowns: Bitcoin -83%, S&P 500 -57%, gold -45%, BET-TR -31% (March 2020), EUR/RON -16%
- Depth and duration matter together

How Long Does Recovery Take?

	MDD	Peak	Trough	Peak regained
S&P 500	−57%	Oct 2007	Mar 2009	Mar 2013 (5.5 years)
BET-TR	−31%	Jan 2020	Mar 2020	Dec 2020 (0.9 years)
Bitcoin	−83%	Dec 2017	Dec 2018	Nov 2020 (3.0 years)
EUR/RON	−16%	Dec 2005	Jul 2007	Jan 2008 (2.0 years)
Gold	−45%	Sep 2011	Dec 2015	Jul 2020 (8.9 years)

- Depth and duration are different risks: gold's drawdown was shallower than Bitcoin's but lasted three times longer
- Fast V-shaped recoveries (BET-TR 2020) versus long underwater periods (gold 2011–2020)
- For EUR/RON a “drawdown” is a fall of the euro against the leu

The Sharpe Ratio

- ▣ **Sharpe ratio** (SR), Sharpe (1966); Sharpe (1994): excess return per unit of volatility

$$SR = \frac{\mu_{\text{ann}} - r_f}{\sigma_{\text{ann}}}$$

- ▶ r_f : risk-free rate; here $r_f = 0$ for comparability across markets
- ▣ **Worked example**, S&P 500 1990–2026
 - ▶ $\mu_{\text{ann}} = 252 \times 0.0331\% = 8.33\%$ (mean log return)
 - ▶ $\sigma_{\text{ann}} = 17.99\%$
 - ▶ $SR = 8.33/17.99 = 0.46$

The Sortino and Calmar Ratios

- **Sortino ratio**, Sortino & van der Meer (1991): penalises only downside deviation

$$SoR = \frac{\mu_{\text{ann}} - r_f}{\sigma_{\text{ann}}^-}, \quad \sigma^- = \sqrt{\mathbb{E}[\min(r_t, 0)^2]}$$

- ▶ σ^- : **downside deviation**, computed from negative returns only
- ▶ S&P 500: $\sigma_{\text{ann}}^- = 12.90\%$, so $SoR = 8.33/12.90 = 0.65$

- **Calmar ratio**: CAGR divided by the absolute maximum drawdown

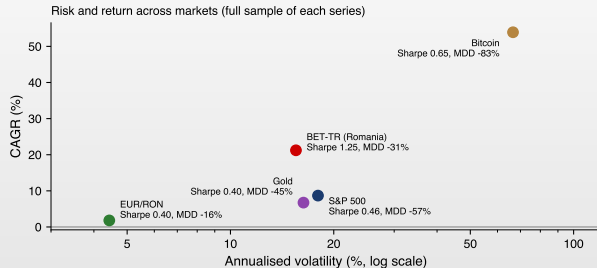
$$\text{Calmar} = \frac{\text{CAGR}}{|MDD|}$$

- ▶ S&P 500: $8.69\%/56.78\% = 0.15$; very sensitive to one extreme episode

Caveats for Performance Measures

- All three ratios are estimates with large sampling error
- Heavy tails and negative skewness make the Sharpe ratio flattering
 - ▶ The Deflated Sharpe Ratio corrects for this and for multiple testing (Chapter 13)
- Different sample periods make comparisons across markets fragile
- Currency and interest rates: BET-TR is in RON, where short-term rates were well above USD rates; with $r_f = 0$ its Sharpe ratio is inflated
- Always report the ratios together with drawdowns and tail statistics

Risk and Return Across Markets



	CAGR	Sortino	Calmar
S&P 500	8.7%	0.65	0.15
BET-TR	21.2%	1.72	0.68
Bitcoin	53.9%	0.91	0.65
EUR/RON	1.8%	0.62	0.12
Gold	6.7%	0.56	0.15

- Different sample periods: not a horse race
- S&P 500 without dividends; BET-TR with dividends
- Higher volatility did not always mean higher Sharpe

Stylised Facts in One Table

Fact	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Heavy tails (JB rejects)	yes	yes	yes	yes	yes
Negative skewness	yes	strong	yes	no	yes
Volatility clustering (ARCH-LM)	yes	yes	yes	yes	yes
Slow decay of ACF $ r_t $	yes	weaker	weaker	yes	yes
Near-zero ACF of r_t	yes	small +	yes	no (+0.17)	yes
Leverage effect ($k = 1$)	-0.12	-0.12	-0.07	+0.10	-0.02

- The core facts (heavy tails, clustering, long memory in volatility) are universal
- Autocorrelation and asymmetry depend on market microstructure and institutions

What a Good Model Must Reproduce

Requirements

- ▣ Time-varying, persistent volatility
- ▣ Heavy-tailed innovations, also after conditioning on volatility
- ▣ Asymmetric response to bad news
- ▣ Approximately uncorrelated returns

- ▣ The random walk with i.i.d. increments following the Normal distribution (Bachelier–Black–Scholes world) violates Facts 2, 3 and 5–11; Fact 4 holds trivially (sums of Normally distributed variables follow the Normal distribution)

Where in this course

- ▣ Ch. 5: GARCH; Ch. 6: multivariate volatility and dependence
- ▣ Ch. 7–8: VaR, Expected Shortfall, extreme value theory; backtesting
- ▣ Ch. 9: realised volatility from intraday data
- ▣ Ch. 13: machine learning must respect the low signal-to-noise ratio

Practical Checklist for Empirical Work

- ▣ **Know your data**
 - ▶ Price vs total-return index; adjusted vs raw prices; fixing vs traded price
 - ▶ Official reference rates where possible (e.g. EUR/RON)
- ▣ **Clean before you model**
 - ▶ Non-trading days (zero volume), holidays, stale quotes, obvious errors
 - ▶ Check extremes: are the largest moves real events?
- ▣ **Use log returns** for single-asset dynamics, simple returns for portfolios
- ▣ **Report robust statistics:** quantiles and tail counts next to kurtosis
- ▣ **Annualise carefully:** 252 vs 365 periods; $\sqrt{P}$ fails under dependence
- ▣ **Reproducibility:** code and data on GitHub, as Quantlets

Key Formulas

Returns

- ▣ $R_t = P_t/P_{t-1} - 1, \quad r_t = \ln(1 + R_t)$
- ▣ $r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$
- ▣ $R_{p,t} = \sum_i w_i R_{i,t}$
- ▣ $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$

Tests

- ▣ $JB = \frac{T}{6} (\hat{S}^2 + (\hat{K} - 3)^2/4)$
- ▣ $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k)$

Volatility

- ▣ $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$
- ▣ Parkinson: $\frac{1}{4 \ln 2} \overline{(h-l)^2}$
- ▣ Mixture: $K = 3 \mathbb{E}[\sigma^4] / \mathbb{E}[\sigma^2]^2$

Performance

- ▣ $SR = (\mu - r_f)/\sigma, \quad \text{SoR} = (\mu - r_f)/\sigma^-$
- ▣ $DD_t = P_t / \max_{s \leq t} P_s - 1$
- ▣ Calmar = $\text{CAGR} / |\text{MDD}|$

Conclusions

- ▣ Returns are **uncorrelated but not independent**: their size is predictable, their sign almost not
- ▣ Heavy tails are the rule: 5σ days are 1000+ times more frequent than the Normal distribution predicts
- ▣ Volatility clusters, has long memory and reacts asymmetrically to losses in equity markets
- ▣ Tails thin out with aggregation, but slowly
- ▣ Crypto shows the same facts with larger amplitude; managed exchange rates and thin markets add autocorrelation
- ▣ Range-based estimators extract more information from OHLC data
- ▣ Risk-adjusted measures must be read together with drawdowns and tail statistics

Further Study on Quantinar

Advanced courses on quantinar.com

-  Statistics of Financial Markets
-  Tukey's g- and h- transformations
-  Applied Time Series Analysis with Python

Code and data

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_01

References (1/3)

- ▣ Bouchaud, J.-P., Matacz, A., Potters, M. (2001). Leverage effect in financial markets: the retarded volatility model. *Physical Review Letters*, 87(22), 228701.
- ▣ Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- ▣ Christie, A.A. (1982). The stochastic behavior of common stock variances: value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.
- ▣ Clark, P.K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1), 135–155.
- ▣ Cong, L.W., Li, X., Tang, K., Yang, Y. (2023). Crypto wash trading. *Management Science*, 69(11), 6427–6454.
- ▣ Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- ▣ Davis, R.A., Mikosch, T. (1998). The sample autocorrelations of heavy-tailed processes with applications to ARCH. *Annals of Statistics*, 26(5), 2049–2080.
- ▣ Ding, Z., Granger, C.W.J., Engle, R.F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106.
- ▣ Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- ▣ Fama, E.F. (1965). The behavior of stock-market prices. *Journal of Business*, 38(1), 34–105.

References (2/3)

- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Garman, M.B., Klass, M.J. (1980). *On the estimation of security price volatilities from historical data*. *Journal of Business*, 53(1), 67–78.
- ▣ Granger, C.W.J., Ding, Z. (1995). *Some properties of absolute return: an alternative measure of risk*. *Annales d'Économie et de Statistique*, 40, 67–91.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance*, 3rd ed. Springer.
- ▣ Härdle, W.K., Harvey, C.R., Reule, R.C.G. (2020). *Understanding cryptocurrencies*. *Journal of Financial Econometrics*, 18(2), 181–208.
- ▣ Hill, B.M. (1975). *A simple general approach to inference about the tail of a distribution*. *Annals of Statistics*, 3(5), 1163–1174.
- ▣ Jarque, C.M., Bera, A.K. (1980). *Efficient tests for normality, homoscedasticity and serial independence of regression residuals*. *Economics Letters*, 6(3), 255–259.
- ▣ Karpoff, J.M. (1987). *The relation between price changes and trading volume: a survey*. *Journal of Financial and Quantitative Analysis*, 22(1), 109–126.
- ▣ Ljung, G.M., Box, G.E.P. (1978). *On a measure of lack of fit in time series models*. *Biometrika*, 65(2), 297–303.
- ▣ Mandelbrot, B. (1963). *The variation of certain speculative prices*. *Journal of Business*, 36(4), 394–419.

References (3/3)

- ▣ Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Praetz, P.D. (1972). The distribution of share price changes. *Journal of Business*, 45(1), 49–55.
- ▣ Rogers, L.C.G., Satchell, S.E. (1991). Estimating variance from high, low and closing prices. *Annals of Applied Probability*, 1(4), 504–512.
- ▣ Sharpe, W.F. (1966). Mutual fund performance. *Journal of Business*, 39(1), 119–138.
- ▣ Sharpe, W.F. (1994). The Sharpe ratio. *Journal of Portfolio Management*, 21(1), 49–58.
- ▣ Sortino, F.A., van der Meer, R. (1991). Downside risk. *Journal of Portfolio Management*, 17(4), 27–31.
- ▣ Taylor, S.J. (1986). Modelling Financial Time Series. Wiley; 2nd ed. (2008), World Scientific, doi:10.1142/6578.
- ▣ Yang, D., Zhang, Q. (2000). Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business*, 73(3), 477–491.
- ▣ Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- ▣ Student (1908). The probable error of a mean. *Biometrika*, 6(1), 1–25.